RAJALAKSHMI ENGINEERING COLLEGE

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B2

Started on	Saturday, 12 September 2026, 2:06 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:07 PM
Time taken	41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c

#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    int dp[n];

    // Read sequence
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        dp[i] = 1;
    }

    // Dynamic Programming
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++) {
            // Non-decreasing: equal values are allowed
            if (a[j] <= a[i]) {
                if (dp[j] + 1 > dp[i]) {
                    dp[i] = dp[j] + 1;
                }
            }
        }
    }

    // Find maximum length
    int ans = dp[0];

    for (int i = 1; i < n; i++) {
        if (dp[i] > ans) {
            ans = dp[i];
        }
    }

    printf("%d", ans);

    return 0;
}


```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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